

Statistics & Probability

ANSWER KEY



Measures of center

- 1 Find the mean of 12, 15, 18, 9, 21.
Sum = 75. Mean = $\frac{75}{5} = 15$.
 - 2 Find the median of 4, 9, 2, 11, 7, 15.
Sorted: 2, 4, 7, 9, 11, 15. Median = $\frac{7+9}{2} = 8$.
 - 3 Find the mode of 3, 7, 5, 7, 2, 7, 9.
The value 7 appears most often, so the mode is 7.
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Measures of spread

- 4 Find the range of the data set 22, 18, 35, 9, 27.
Range = $35 - 9 = 26$.
 - 5 Find the interquartile range (IQR) of 4, 7, 9, 12, 15, 18, 21.
Lower half (excluding the median 12): 4, 7, 9, so $Q_1 = 7$. Upper half: 15, 18, 21, so $Q_3 = 18$. IQR = $18 - 7 = 11$.
 - 6 A data set has a range of 40 and a minimum value of 15. Find the maximum value.
Maximum = $15 + 40 = 55$.
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Basic probability

- 7 A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.
Total marbles = 10. $P(\text{green}) = \frac{2}{10} = \frac{1}{5}$.
 - 8 A fair six-sided die is rolled once. Find $P(\text{even number})$.
Even outcomes: {2, 4, 6}. $P = \frac{3}{6} = \frac{1}{2}$.
 - 9 A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than 5})$.
Numbers greater than 5: {6, 7, 8}. $P = \frac{3}{8}$.
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Combined probability: AND / OR

- 10 A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.
4 kings + 4 queens = 8 favorable cards. $P = \frac{8}{52} = \frac{2}{13}$.

- 11 Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.
 $P(A \cap B) = 0.4 \times 0.5 = 0.20$.
- 12 A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.
 $P = \frac{6}{10} \times \frac{6}{10} = \frac{36}{100} = \frac{9}{25}$.
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Probability without replacement

- 13 A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.
 $P = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \frac{5}{33}$.
- 14 A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.
 $P = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$.
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Data displays

- 15 The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.
Experimental $P(4) = \frac{9}{40} = 0.225$. Theoretical $P(4) = \frac{1}{6} \approx 0.167$. The experimental probability is somewhat higher than the theoretical probability, which is expected random variation in a finite sample.
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Expected value

- 16 A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.
Expected winnings $= 0.1 \times 10 = 1$ dollar. Expected net profit $= 1 - 2 = -1$ dollar (an expected loss of 1 dollar per play).
- 17 A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.
 $E = \frac{5 + 10 + 15 + 20}{4} = \frac{50}{4} = 12.5$ points.
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Effects of transformations on statistics

- 18 A data set has mean 20. If every value in the set is increased by 5, find the new mean.
New mean $= 20 + 5 = 25$.

- 19 A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.
New standard deviation $= 3 \times 4 = 12$.
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A well-designed word problem: a survey of students

- 20 This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.

(a) By inclusion-exclusion, the number doing at least one activity is $120 + 80 - 30 = 170$. Neither $= 200 - 170 = 30$. (b) $P(\text{sport or instrument}) = \frac{170}{200} = 0.85$. (c) $P(\text{instrument} \mid \text{sport}) = \frac{P(\text{both})}{P(\text{sport})} = \frac{30}{120} = \frac{1}{4}$.